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ABSTRACT

In disc golf, having a comfortable and proper grip on the disc is crucial to achieving optimal throw dynamics, and the placement of the thumb is a key component of the grip. This research investigated the relationship between thumb position on a mid-range disc and the resulting angular speed, translational speed, and torque generated during the throw. Using a Discraft Buzzz disc equipped with a TechDisc sensor, measurements were obtained for throws made using five different thumb positions, starting near the outer edge of the disc ($p = 1$) and ending near the center of the disc ($p = 5$). Participants were sorted into two skill level groups that were determined based on their Professional Disc Golf Association rating. Each participant performed five throws at each thumb position with maximum effort, and angular and translational speeds of each throw were measured. The results varied between skill groups, but collective data displayed a general linear correlation between angular and translational speeds achieved across all thumb positions. While angular speed and applied torque were observed to be maximum for the thumb position with the largest radial distance from the center of the disc ($p = 1$), thumb positions located ~ 3 – 5 cm radially inward from the outer edge ($p = 2$ and $p = 3$) resulted in the highest translational speeds. In general, a universally favorable thumb position was determined as that resulting in simultaneously large values of both angular and translational speeds for all participants, which occurred closest to the $p = 2$ position (~ 3 cm from the outer edge).

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I. INTRODUCTION

Disc golf is a relatively young sport that has experienced consistent growth since its inception, and the disc golf community has significantly grown in popularity over the past several years. The sport was invented based on various flying toy discs contrived in the mid-twentieth century. The idea evolved until Wham-O produced a Pro Model disc in 1964,¹ and disc golf officially became a professional sport in 1976.² As of 2022, the Professional Disc Golf Association (PDGA) reported a total of 130 700 registered members, which represents 19% and 84% increases over 1 and 2 year spans, respectively, with over 10 000 courses existing worldwide and nearly 9000 sanctioned events held throughout the year.³ Even with the growing popularity, there is a general lack of research devoted to the mechanics of disc golf throws.

Compared to catch-style discs, disc golf discs typically have smaller diameters, higher densities, and a more aerodynamic outer rim. In disc golf, there are three main disc types, namely, drivers, mid-ranges, and putters. Driver discs are typically very thin and

sharp-edged to reduce the amount of drag acting on the disc while in flight and can travel greater than 500 feet if thrown with high enough speeds. Putters are usually the thickest of the three disc types, experience higher drag forces than drivers, and are commonly used when players are close to the basket. Finally, mid-range discs typically share characteristics of both drivers and putters, which offer players a reliable option for shots that require both control and accuracy and are typically used for shots between 200 and 300 feet of distance.

In general, there are two commonly used disc golf throwing styles: backhand (where the disc is pulled across the body from the player's non-dominant side toward their dominant side before release) and forehand (where the disc is thrown sidearm with the player's dominant hand). While there are currently no accessible statistical data that report which style is most prevalent in the disc golf community, the backhand throw is widely assumed to be the most common based on various instructional resources and media coverage of professional disc golf tournaments.⁴ No matter which style or type of disc a player is throwing, a comfortable and proper grip

is essential to achieving accurate and appropriate distance for each shot on the course. There are different ways for players to position their fingers while gripping the disc, such as a fan grip (where the fingers are spread out near the inner edge of the disc), a stacked grip (where the fingers are stacked on top of one another along the inner edge of the disc), or a power grip (which is the most common and involves tightly wrapping all four fingers around the rim of the disc). The positioning of the thumb on the top side of the disc is another aspect of any disc golf grip. The thumb's position is crucial, because it counterbalances the fingers underneath to secure the disc in the hand and also determines characteristics related to applied torque on the disc during a throw.

The goal of this study is to investigate the effects of thumb position implemented via a backhand power grip on the resulting angular speed (or spin rate), translational speed (or launch speed), and torque generated on a mid-range disc. In the experiment, participants were split into two different skill level groups based on each participant's PDGA rating, where group I included participants with ratings less than 900 (or unrated) and group II included participants with ratings of 900 or higher. Each participant performed repeated backhand throws using a Discraft Buzzz mid-range disc equipped with a TechDisc flight sensor utilizing five different thumb positions. Each participant was instructed to employ a consistent throwing motion to the best of their ability while only changing the position of the thumb, and angular and translational speeds of the disc were measured for each throw. One previous study was conducted where the trajectories of different disc shapes were tested computationally by combining computational fluid dynamics and rigid body dynamics,⁵ and two Ph.D. dissertations have investigated general throw biomechanics.^{1,6} However, this is the first instance in the literature of an experimental study performed to investigate the effects of thumb position on the resulting launch characteristics achieved during a disc golf throw.

II. THEORY

The angular momentum ($\vec{L}$) imparted to a spinning disc is given by

$$\vec{L} = I\vec{\omega}, \quad (1)$$

where I is the moment of inertia and $\vec{\omega}$ is the angular velocity of the disc. Newton's second law of rotational motion states that

$$\vec{\tau} = \frac{d\vec{L}}{dt}, \quad (2)$$

where $\vec{\tau}$ is the net torque and $\frac{d\vec{L}}{dt}$ is the first derivative of angular momentum with respect to time. At the release point of the throw, the center of the disc was considered to be the axis of rotation, and all thumb positions were measured relative to this axis. For all measurements of torques applied by participants during throws in this study, the angular velocity and angular momentum vectors were assumed to be directed purely in the direction perpendicular to the plane of the disc, which was defined to be along the $\hat{z}$ -axis. As a disc is thrown with higher angular speeds, the angular momentum of the disc increases [according to Eq. (1)]. Discs that impart larger values of angular momentum upon launch will require larger values of net torque to cause noticeable changes in the direction of the disc's axis of rotation while in flight. Thus, disc golf throws with higher angular

speeds will possess higher degrees of gyroscopic stability than throws with lower angular speeds. Gyroscopic stability can be described as a disc's tendency to resist changes to its axis of rotation due to conservation of angular momentum. Increased gyroscopic stability is favorable for disc throws that require accuracy and control due to the larger external torques (primarily due to aerodynamic forces) that are required to change the disc's rotational state during flight. This concept motivates the use of angular speed measurements to partially determine the optimal thumb position in disc golf throws.

When using a power grip, forces applied to the disc due to the grip include the static force of friction between the thumb and the top of the disc, as well as the force applied by the fingers on the inner rim of the disc. During a throw, there may be additional force contributions applied to the disc that are not purely perpendicular to the radial direction. For example, the downward force of the thumb on the top of the disc may cause the disc material to flex in such a way that a component of the force is applied along the radial direction toward the outer rim. However, since such a force would not directly contribute to the applied torque on the disc, this study utilized a simplified model that assumes a net horizontal force that is applied perpendicular to the radial direction. The net horizontal force ($\vec{F}_{hor}$) is assumed to be the sum of the frictional force and the horizontal component of the force from the fingers on the inner rim and generates a net torque on the disc, given by

$$\vec{\tau} = \vec{r} \times \vec{F}_{hor}, \quad (3)$$

where $\vec{\tau}$ is the net torque applied to the disc and $\vec{r}$ is the radial displacement of the thumb position relative to the center of the disc. A basic schematic illustrating the general layout of the $\vec{F}_{hor}$ and $\vec{r}$ vectors during a disc golf throw is shown in Fig. 1. The grip depicted in Fig. 1 represents the general orientation of the hand during the accelerating phase of the throw, with the translational velocity directed in the same direction as $\vec{F}_{hor}$. Upon release, the player's elbow and wrist will hinge just before the grip force is removed, setting the disc into flight.

From the definition of a cross product, consequently,

$$\tau = rF_{hor} \sin \theta_p, \quad (4)$$

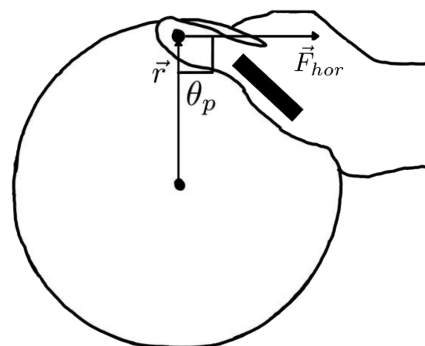


FIG. 1. Schematic depicting the general orientation of the horizontal net force ($\vec{F}_{hor}$) and radial displacement ($\vec{r}$) vectors when using a power grip. The black rectangle illustrates the general location of the fingers located on the underside of the disc.

where τ , r , and F_{hor} are the respective magnitudes of $\vec{\tau}$, $\vec{r}$, and $\vec{F}_{hor}$, and θ_p is the angle between $\vec{r}$ and $\vec{F}_{hor}$ at each thumb position (p).

The radial displacement vector of the thumb and the friction force vector were assumed to be orthogonal ($\theta_p \approx 90^\circ$ and $\sin \theta_p \approx 1$) for all thumb positions. In addition, each participant was asked to consistently throw the disc with full power, so F_{hor} was assumed to remain constant for each participant's set of throws (regarding applied torque measurements). Thus, the magnitude of the applied torque was assumed to depend on the thumb position via the corresponding radial displacement. As shown in Fig. 2, the selected thumb positions began near the outer edge of the disc ($p = 1$) with sequential positions approaching the center of the disc. The fifth thumb position ($p = 5$) was located near the center of the disc with a radial displacement of $r = 2.61 \pm 0.05$ cm. The radial displacement of the thumb (relative to the center of the disc) decreased as thumb positions increased from $p = 1$ to $p = 5$. According to Eq. (4), the magnitude of the net torque applied to the disc was expected to decrease as r was decreased (and p was increased).

An alternate form of Newton's second law of rotational motion states that

$$\vec{\tau} = \mathbf{I} \cdot \vec{\alpha}, \quad (5)$$

where $\vec{\alpha}$ is the disc's angular acceleration vector and $\mathbf{I}$ is the disc's inertia tensor for rotations about the vertical axis through its center. The inertia tensor is given by

$$\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}, \quad (6)$$

where I_{xx} , I_{yy} , and I_{zz} represent the disc's moments of inertia about the x , y , and z -axes respectively, and the off-diagonal elements are

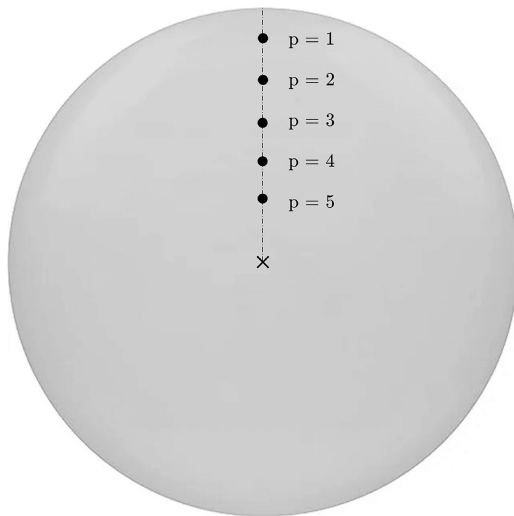


FIG. 2. Schematic showing the top view of the mid-range disc with five marked thumb positions (p) at various radial displacements from the central axis of rotation.

the six products of inertia for rotations about all three coordinate axes. Each moment of inertia of the disc is given by

$$I_{ii} = \int_V \rho(j^2 + k^2) dV, \quad (7)$$

where ρ is the mass density of the disc and V is the disc's volume. Each product of inertia of the disc is given by

$$I_{ij} = \int_V \rho(ij) dV, \quad (8)$$

where i , j , and k represent x , y , and z in some arbitrary order.⁷ The coordinate axes were chosen such that the x -axis and y -axis lie in the plane of the disc (and rotate along with the disc), and the z -axis points upward through the disc's axis of rotation (with the origin at the center of the disc). Thus, $\mathbf{I}$ is diagonal due to rotational symmetry about the z -axis. Assuming a constant applied torque throughout the period of angular acceleration, in addition to angular velocity and angular acceleration vectors purely directed along the z -axis, it follows from Eq. (5) that

$$\tau = I_{zz} \left(\frac{\omega}{t} \right), \quad (9)$$

where ω is the angular speed of the disc about the z -axis (upon release) and t is the duration of angular acceleration during the throw. Combining Eqs. (4) and (9) results in an expression for the angular speed of the disc, given by

$$\omega = \frac{r F_{hor} t}{I_{zz}}, \quad (10)$$

which shows a predicted direct relationship between angular speed (ω) and radial displacement (r) of the thumb position.

III. EXPERIMENTAL METHODS

The experimental setup consisted of a Discraft Buzzz mid-range disc equipped with a TechDisc flight sensor, a 7×7 foot throw net, and the TechDisc data collection software. The angular and translational speeds of backhand power grip throws were measured for a total of 24 participants, including 15 participants in group I and nine participants in group II. Each participant was instructed to throw five replicate throws using each marked thumb position at full power, resulting in a total of 25 throws per participant and 600 total throws analyzed across the study. Paper tabs were adhered onto the top face of the disc to clearly mark the five thumb positions (shown in Fig. 2), and participants were instructed to use a consistent throwing motion and a power grip for all throws. Participants were instructed to perform their preferred starting walkup routine before making their throws. Once a participant finished all five sets of throws, the corresponding data were sorted into either group I or group II depending on the PDGA rating⁸ of the participant. Participants without a PDGA rating and those with ratings <900 were placed in the novice group (group I), while participants with PDGA ratings ≥ 900 were placed in the advanced group (group II). These groupings were created to observe any differences in trends observed for beginner/novice players and more experienced or professional players.

TABLE I. Reported measured values of average angular speed (ω) and average translational speed (v) with respective uncertainties (σ_ω and σ_v). Data are reported for each thumb position (p) and radial distance from the center of the disc (r), as well as for each individual skill group and for both groups combined. Note that the group label "C" denotes the combined skill groups.

p	r (± 0.05 cm)	Group	ω (rpm)	σ_ω (rpm)	v (mph)	σ_v (mph)
1	9.40	I	796	17	45.1	0.7
2	7.60	I	774	17	46.6	0.7
3	5.80	I	739	19	46.1	0.8
4	4.20	I	738	21	44.6	0.7
5	2.61	I	739	21	44.8	0.8
1	9.40	II	1149	38	55.4	1.5
2	7.60	II	1145	31	57.6	1.2
3	5.80	II	1120	31	58.3	1.0
4	4.20	II	1107	27	56.8	0.9
5	2.61	II	1090	27	56.1	1.2
1	9.40	C	900	24	48.1	0.8
2	7.60	C	890	23	50.0	0.8
3	5.80	C	861	25	50.0	0.9
4	4.20	C	852	24	48.4	0.8
5	2.61	C	856	24	48.6	0.9

The radial displacements for each thumb position ranged from $r = 9.40$ cm ($p = 1$) to $r = 2.61$ cm for $p = 5$ (relative to the center of the disc). All measurements related to radial displacements for each thumb position can be found in Table I. To determine the disc's moment of inertia for rotations about the z -axis (I_{zz}), the duration of the accelerating phase of a throw was determined via video analysis of ten replicate throws. Assuming uniform density (ρ), the mass of the disc ($m = 175$ g) and other relevant physical measurements of the mid-range disc (shown in Fig. 3) were used via Eq. (7) to determine I_{zz} as 12.9 ± 0.4 kg cm².

IV. RESULTS AND ANALYSIS

Figure 4 shows the average angular speed measurements plotted as a function of radial distance of the thumb position to observe general trends relating the thumb position and spin rates generated during the throw. In addition, Fig. 5 shows the average translational speed measurements plotted as a function of radial distance

to observe effects of the thumb position on linear launch speed. As a means of displaying the distribution of measured quantities observed for each thumb position throughout the study, histograms of all angular and translation speed measurements for each thumb position are presented in the [supplementary material](#) document. To assess the normality of each angular and translational speed dataset for each thumb position, the Shapiro–Wilk test was performed for each dataset via Maxima. Corresponding p -values (denoted by the symbol P) are displayed in Fig. S1 and indicate the likelihood that the dataset is Gaussian (where $P < 0.05$ suggests that the dataset is not normally distributed). Since the data collected in this study were obtained from 24 different participants, the resulting datasets were not expected to produce normal distributions due to the inherent variations in throwing motion, experience, and ability (which is manifested in the rather broad error bars corresponding to measured averages for each dataset). Increasing the number of participants and replicate throws per participant would likely cause the datasets to approach more normal distributions and result in average measured values with more narrow confidence intervals.

For applied torque measurements, the average duration of the accelerating phase of a throw was measured to be 0.27 ± 0.01 s. With measured values of t , ω , and I_{zz} , Eq. (9) was used to determine the magnitude of the applied torque on the disc at each thumb position for the combined groups. As shown in Fig. 6, these values of applied torque were plotted as a function of radial distance of the thumb position. Corresponding uncertainties for measured torques were determined using the general propagation of error equation⁹ applied to Eq. (9).

In addition to looking at direct effects of the thumb position on spin rates and launch speeds, translational speed was plotted as a function of angular speed for group I, group II, and all participants (see Figs. 7–9, respectively). Interestingly, linear trends were observed for each individual group and the combined groups, showing a positive linear correlation between translational and angular speeds generated during disc golf throws. Average angular and translational speeds of the disc thrown at each thumb position for individual and combined skill groups, along with the corresponding uncertainties for each measurement (determined via standard error of the mean), are reported in Table I.

V. DISCUSSION

Figure 4(a) shows that the average values of angular speed were ~50% higher for the advanced group (group II) than the novice

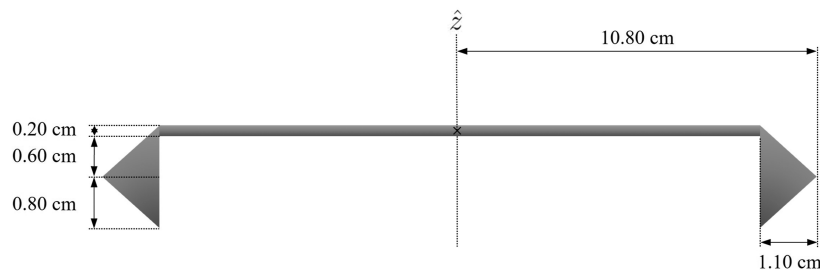


FIG. 3. Cross-sectional view of the simplified model of the mid-range disc used to determine the disc's moment of inertia for rotations about the z -axis (I_{zz}).

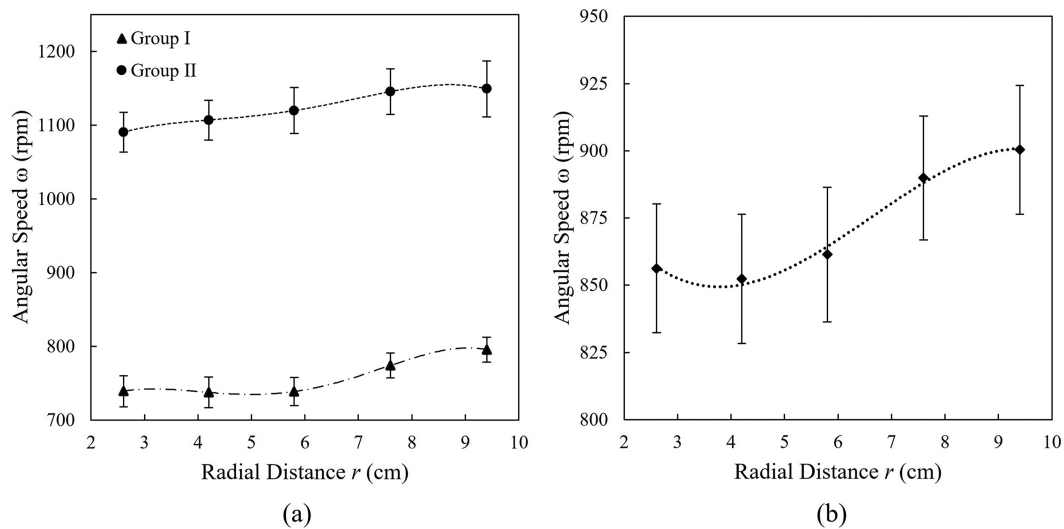


FIG. 4. Plots of the average angular speed of throws as a function of radial distance of the thumb position relative to the center of the disc (a) for each skill group and (b) for the combined groups. For ease of viewing, polynomial fits were applied to the data on each plot.

group (group I) for each thumb position. For the novice group, the maximum average value of angular speed was observed for the thumb position with the largest radial distance from the center of the disc ($p = 1$). For the advanced group, average angular speeds showed a consistent increase as the radial distance of the thumb position was increased, with a maximum value observed at $p = 1$. The combined group data shown in Fig. 4(b) also displayed a maximum value of angular speed at the $p = 1$ thumb position. These results suggest that thumb positions further away from the center

of the disc generally result in higher spin rates in mid-range disc golf throws.

Figure 5(a) shows that the average values of translational speed were $\sim 25\%$ higher for the advanced group (group II) than the novice group (group I) for each thumb position. For the novice group, the maximum average value of translational speed was observed for the $p = 2$ thumb position. For the advanced group, the maximum value of translational speed was observed for the $p = 3$ thumb position. For the combined groups, the maximum value of translational speed was

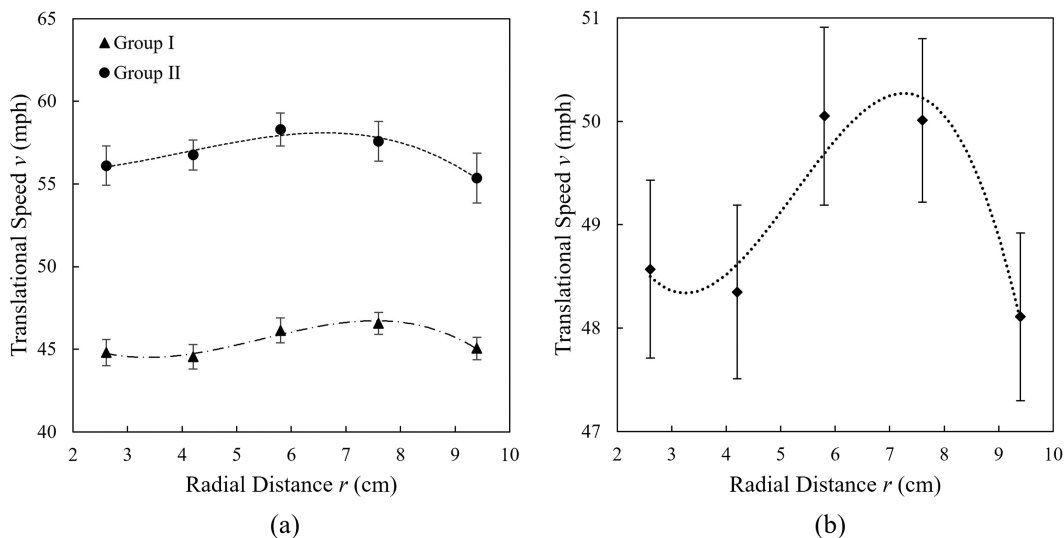


FIG. 5. Plots of the average translational speed of throws as a function of radial distance of the thumb position relative to the center of the disc (a) for each skill group and (b) for the combined groups. For ease of viewing, polynomial fits were applied to the data on each plot.

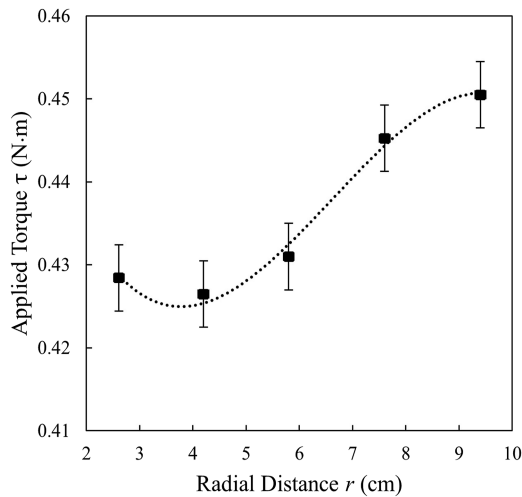


FIG. 6. Plot of the magnitude of applied torque as a function of radial distance of the thumb position relative to the center of the disc for the combined skill groups. For ease of viewing, a polynomial fit was applied to the data.

observed for the $p = 2$ and $p = 3$ thumb positions, with less than 0.1% difference between their average values [see Fig. 5(b)]. However, the variance in translational speeds achieved for all participants was slightly lower for the $p = 2$ position than the $p = 3$ position. One possible reason for the higher translational speeds achieved for the $p = 2$ and $p = 3$ positions over the $p = 1$ position is the increased ability to flex the disc surface with the thumb at the $p = 2$ and $p = 3$ positions. This flexion likely produces greater frictional forces during the throw, which would result in increased translational accelerations and speeds.

Figure 6 shows the applied torque measurements plotted as a function of radial distance of the thumb position, displaying a nearly identical trend to that of the angular speed vs radial distance plot in Fig. 4(b) (as expected). Measured values ranged from 0.42 to 0.45 N m, and a maximum value was observed for the $p = 1$ thumb position. The similarity in trends between angular speed and applied torque when plotted as a function of radial distance is consistent with the linear dependence between applied torque and angular speed predicted by Eq. (9). However, average values of angular speed and applied torque for the $p = 5$ thumb position displayed an unexpected increase in the combined group data, which was inconsistent with the trend predicted by Eqs. (9) and (10). This increase is likely due to grip variations caused by the general discomfort of throws performed at the $p = 5$ thumb position. These grip variations may have also caused changes in F_{hor} , $\vec{r}$, θ_p , and/or general mechanics of the throw that resulted in slight increases in torque and angular speed relative to those achieved at the $p = 4$ position.

Perhaps the most general observation among data collected in this study was the positive linear correlation observed in plots of translation speed vs angular speed for individual and combined skill groups (see Figs. 7–9). The novice group displayed a lower degree of correlation ($R^2 = 0.4771$) than the advanced group ($R^2 = 0.6253$), likely due to inconsistencies in throwing mechanics for the novice group. However, the combined group data showed a high linear correlation of $R^2 = 0.7404$, providing convincing support for the linear correlation between the launch speed and spin rate in mid-range disc golf throws.

In addition to the high variability of individual throwing styles, throw power, and throw repeatability, there were also several assumptions and details surrounding experimental design that may have led to increased contributions of random error. It was assumed that each participant would complete each throw with their thumb placed at the appropriate and precise thumb position throughout the

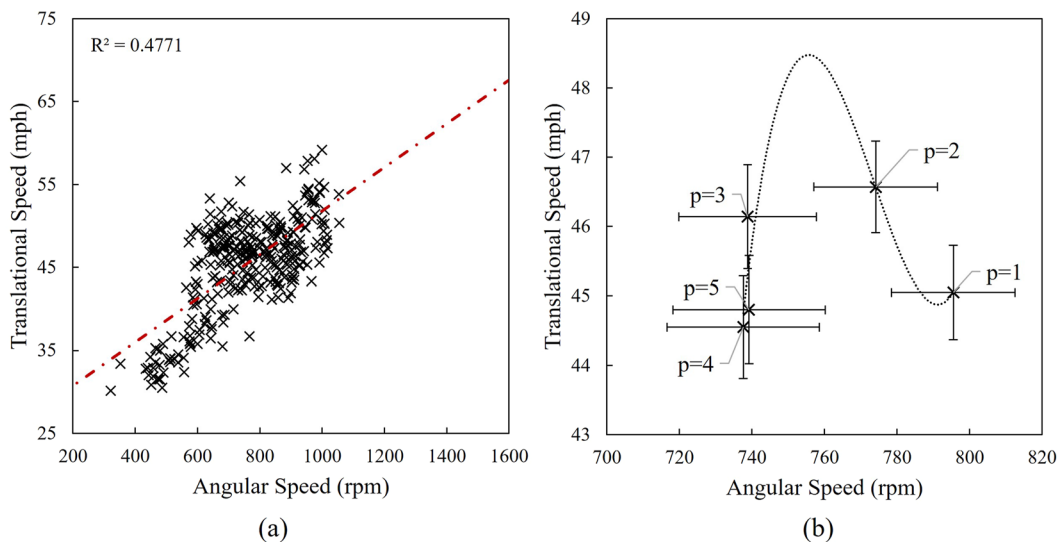


FIG. 7. (a) Plot of angular speed vs translational speed for all participant throws in group I with a linear fit and displayed R^2 value and (b) plot of average angular speed vs average translational speed for each thumb position (p) for group I (with a polynomial fit applied for clarity).

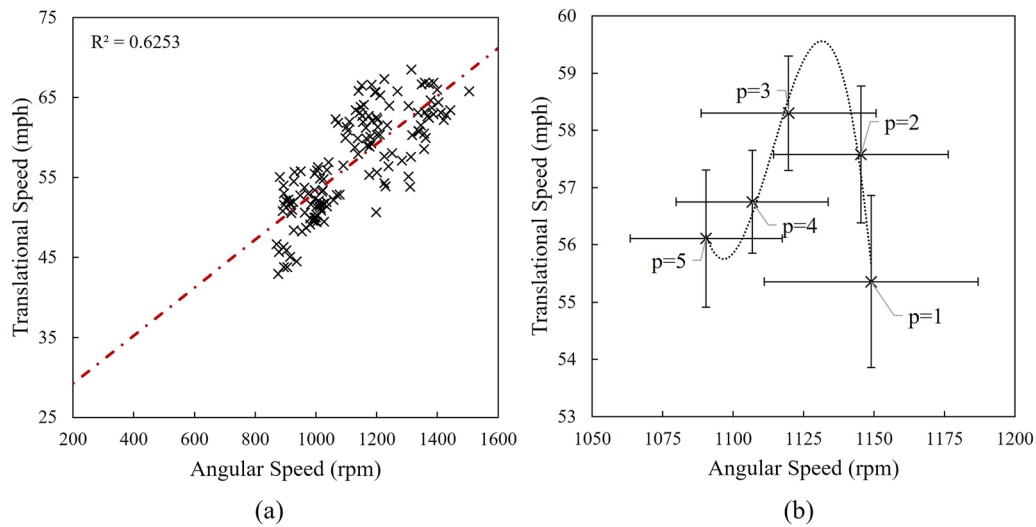


FIG. 8. (a) Plot of angular speed vs translational speed for all participant throws in group II with a linear fit and displayed R^2 value and (b) plot of average angular speed vs average translational speed for each thumb position (p) for group II (with a polynomial fit applied for clarity).

duration of each throw. However, it is unlikely that this assumption held for every recorded throw, which would increase random error contributions. For applied torque measurements, it was assumed that the magnitude of the force of friction between the thumb and the surface of the disc remained constant across all throws since participants were instructed to perform each throw with maximum effort. This is also unlikely to have occurred as physical fatigue related to repeated high effort throws and the absence of a warm-up or stretching routine may have resulted in participants being unable

to throw at full power for all 25 throws. In addition, it was assumed that the duration of angular acceleration on the disc was constant for all participants and throws, but it is likely that this time varied between throwers due to a wide variety of throwing motions.

When average values for each group were plotted for each thumb position, optimal thumb positions were determined by observing which positions simultaneously resulted in the largest values of both angular and translational speeds. Maximizing both of these values should result in the optimal combination of launch

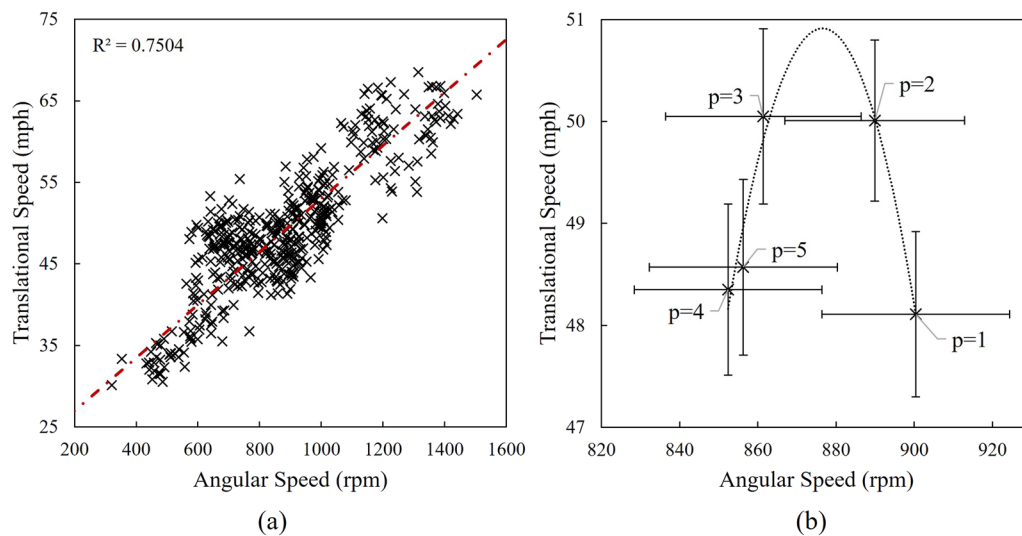


FIG. 9. (a) Plot of angular speed vs translational speed for all participant throws across both skill groups with a linear fit and displayed R^2 value and (b) plot of average angular speed vs average translational speed for each thumb position (p) for combined groups (with a polynomial fit applied for clarity).

speed and spin rate, which promotes shots with a balance of high gyroscopic stability and distance. For the novice group, Fig. 7(b) shows that the $p = 2$ position resulted in the largest value of translational speed, while the $p = 1$ position achieved the maximum average angular speed. While it is unclear whether maximizing angular or translational speed in a disc golf throw leads to higher performance, results suggest that the optimal thumb position for the novice group is most likely located near or between the $p = 1$ and $p = 2$ positions. For the advanced group, Fig. 8(b) shows that the $p = 3$ position resulted in the largest value of translational speed. While the $p = 1$ position again resulted in the highest value of angular speed, the $p = 2$ position achieved a similar average angular speed to $p = 1$ but resulted in a 4% increase in translational speed relative to the $p = 1$ position. Thus, the optimal thumb position for the advanced group is most likely located near or between the $p = 2$ and $p = 3$ positions. When the groups were combined, Fig. 9(b) shows that the $p = 2$ position displayed nearly maximum values for both translational and angular speeds relative to the other thumb positions. The collective dataset provides convincing support for the $p = 2$ thumb position to serve as a universally favorable position for achieving throws with simultaneously large spin rates and launch speeds.

VI. CONCLUSIONS

This study investigated the relationship between the thumb position on a mid-range disc and the resulting angular speed, translational speed, and applied torque achieved in disc golf throws. Data were collected for novice and advanced skill groups, and results were analyzed to determine the optimal thumb position for maximizing the spin rate and launch speed. All throw data collected in this study (for individual and combined group sets) displayed a linear correlation between angular speed and translational speed generated during throws. These results suggest that disc golf throws generating higher spin rates generally result in higher launch speeds.

For the novice group (group I), angular speed was maximum for the thumb position with the largest radial distance from the center of the disc ($p = 1$). However, translational speeds showed a noticeable decrease at this thumb position. Favorable thumb positions for the novice group were deemed to be near or between the $p = 1$ and $p = 2$ positions due to high simultaneous average values of angular and translational speeds. For the advanced group (group II), favorable thumb positions were located near or between the $p = 2$ and $p = 3$ positions.

For both groups combined (group C), applied torque was measured, and values ranged from 0.42 to 0.45 N·m. A maximum value of applied torque was observed for the $p = 1$ thumb position, which followed the same trend observed for the angular speed data. However, simultaneously large values of angular and translational speeds were observed at or near the $p = 2$ position for the combined group, which suggests that this position may be universally favorable for achieving disc golf throws with high spin rates and launch speeds. In general, the $p = 1$, $p = 2$, and $p = 3$ thumb positions universally outperformed the $p = 4$ and $p = 5$ positions in terms of resulting angular and translational speeds. Thumb positions located at the largest radial distances from the center resulted in the highest spin rates, while thumb positions located ~3–5 cm radially inward from the outer edge resulted in the highest launch speeds. More research needs to be performed to determine the relative roles of angular

speed vs translational speed regarding specific characteristics of disc golf throws (such as overall distance, accuracy, or shot shape). The results reported in this study may not be universal and could likely be different for other disc types with varying physical dimensions, disc shapes, and mass distributions.

While this work represents the first instance in the literature of an experimental study dedicated to investigations of disc golf throw mechanics related to thumb position, future work needs to be performed to determine whether these results are valid for other disc types (such as drivers or putters), grips, and other parameters of interest (such as total distance and accuracy of throws).

SUPPLEMENTARY MATERIAL

Histograms including all data obtained in this study for each skill group and thumb position are included in the [supplementary material](#) document. Shapiro–Wilk tests were performed for each dataset, and corresponding p -values (P) and thumb positions (p) are reported on each plot.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

Informed consent was obtained from all study participants, and IRB protocol number 2023-24-47 was approved by the Berry College Institutional Review Board for Human Subjects Research.

Author Contributions

N.K. contributed to experimental design, data collection, analysis/interpretation of results, and manuscript preparation. H.M. contributed to experimental design, data collection, analysis/interpretation of results, and manuscript preparation. C.C. contributed to experimental design, data collection, analysis/interpretation of results, and manuscript preparation. Z.L. contributed to study conception, data collection, analysis/interpretation of results, and manuscript preparation. All authors reviewed the results and approved the final version of the manuscript.

Noah Koch: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (lead); Supervision (lead); Writing – original draft (lead); Writing – review & editing (supporting). **Hayden McGuire:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (supporting). **Connor Cole:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (supporting); Writing – review & editing (supporting). **Zachary Lindsey:** Conceptualization (lead); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (lead); Supervision (lead); Writing – original draft (supporting); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#) and from the corresponding author upon reasonable request.

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